

Lecture 02

Vectors, Linear Combinations, Span, and Feature Spaces

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Lecture Focus

Understanding vectors as data representations, combining vectors, describing the space they generate, and applying vector operations to features, embeddings, similarity, and prediction.

[\[Data Eng. Link: Feature Engineering, Data Representation\]](#)

1 Lecture Overview

A vector is more than a list of numbers. It can represent an observation, a set of model parameters, a direction, or an embedding. Machine-learning systems convert complex objects such as customers, images, and words into vectors so that mathematical operations can be applied to them.

This lecture develops the concepts of vector operations, linear combinations, span, feature spaces, vector length, distance, and similarity.

Learning Outcomes

After completing this lecture, students should be able to:

- Represent real-world observations as vectors.
- Perform vector addition, subtraction, and scalar multiplication.
- Calculate dot products, vector norms, and Euclidean distances.
- Construct and interpret linear combinations of vectors.
- Explain the span of a set of vectors.
- Interpret a dataset as points in a feature space.
- Compute cosine similarity between vector representations.
- Implement vector operations using Python and NumPy.

Vectors in IT Applications

IT Application	Vector Representation
Tabular prediction (e.g., student grades)	Age, GPA, attendance, etc.
Image recognition	Flattened or learned pixel features
Natural-language processing	Word, sentence, or document embeddings
Recommendation systems	User-preference and item-feature vectors
Neural networks	Inputs, weights, activations, and gradients
Clustering (e.g., customer segmentation)	Points compared using distance or similarity

2 Vectors and Their Structure

A vector in $\mathbb{R}^n$ is an ordered list of n real numbers:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

2.1 Column and Row Vectors

A column vector has shape $n \times 1$, while a row vector has shape $1 \times n$. The transpose symbol T changes rows into columns and columns into rows.

2.2 Vector Operations

For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and scalar c :

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}, \quad c\mathbf{x} = \begin{bmatrix} cx_1 \\ \vdots \\ cx_n \end{bmatrix}.$$

3 Length, Direction, and Distance

3.1 Euclidean Norm

$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

3.2 Unit Vector and Normalization

A non-zero vector can be normalized by $\tilde{\mathbf{x}} = \mathbf{x} / \|\mathbf{x}\|_2$.

3.3 Euclidean Distance

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2.$$

Distance is used in nearest-neighbour classification, clustering, anomaly detection, and similarity-based retrieval. [\[Data Eng. Link: Data Pipelines, Data Cleaning\]](#)

3.4 Feature Scaling Matters

Suppose a student vector contains age and annual family income. Income may have values in hundreds of thousands, while age may have values near 20. Without scaling, income will dominate Euclidean distance.

4 Dot Product and Similarity

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the dot product is:

$$\mathbf{x}^T \mathbf{y} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n.$$

4.1 Geometric Form

$$\mathbf{x}^T \mathbf{y} = \|\mathbf{x}\|_2 \|\mathbf{y}\|_2 \cos \theta,$$

where θ is the angle between the vectors.

4.2 Cosine Similarity

$$\text{cosSim}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x}^T \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}.$$

Recommendation Example for IT Students

Let a user's interests in technology, mathematics, and design be $\mathbf{u} = [5, 4, 1]^T$. Two course profiles are $\mathbf{c}_1 = [5, 5, 1]^T$ and $\mathbf{c}_2 = [1, 2, 5]^T$. Cosine similarity can be used to identify which course profile is better aligned with the user's interests. [\[Data Eng. Link: Recommendation Systems\]](#)

5 Linear Combinations

A linear combination of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ is any vector of the form:

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k,$$

where $c_1, \dots, c_k$ are scalars.

5.1 Weighted Prediction as a Linear Combination

Suppose normalized attendance, assignment performance, and quiz performance are $\mathbf{x} = [0.80, 0.70, 0.90]^T$, and the weights are $\mathbf{w} = [0.40, 0.30, 0.30]^T$. The weighted score is $\mathbf{w}^T \mathbf{x} = 0.80$. This score is a linear combination of the student's features.

5.2 Matrix-Vector Form

For several observations stored as rows of X , the predictions can be calculated simultaneously: $\hat{\mathbf{y}} = X\mathbf{w}$. Each entry of $\hat{\mathbf{y}}$ is the dot product of one observation with the weight vector.

6 Span and Feature Spaces

The span of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ is the set of all their linear combinations:

$$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \{c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k : c_i \in \mathbb{R}\}.$$

Feature Space

If a dataset has n features, every observation is represented as a point in an n -dimensional feature space. [\[Data Eng. Link: Data Visualization, Dimensionality Reduction\]](#)

7 Python and NumPy Practice

7.1 Vector Operations

```

1 import numpy as np
2 x = np.array([2.0, 1.0])
3 y = np.array([-1.0, 3.0])
4 print("x + y:", x + y)
5 print("2x - y:", 2 * x - y)
6 print("Dot product:", x @ y)
7 print("Length of x:", np.linalg.norm(x))
8 print("Distance:", np.linalg.norm(x - y))

```

7.2 Normalization and Cosine Similarity

```

1 def normalize(v):
2     length = np.linalg.norm(v)
3     if np.isclose(length, 0):
4         raise ValueError("Zero vector cannot be normalized.")
5     return v / length
6
7 def cosine_similarity(a, b):
8     return (a @ b) / (np.linalg.norm(a) * np.linalg.norm(b))
9
10 u = np.array([5.0, 4.0, 1.0])
11 c1 = np.array([5.0, 5.0, 1.0])
12 c2 = np.array([1.0, 2.0, 5.0])
13 print("Normalized user:", normalize(u))
14 print("Similarity with course 1:", cosine_similarity(u, c1))
15 print("Similarity with course 2:", cosine_similarity(u, c2))

```

8 Review and Independent Practice

Practice Tasks

1. For $\mathbf{x} = [2, -1, 3]^T$ and $\mathbf{y} = [1, 4, 0]^T$, calculate $\mathbf{x} + \mathbf{y}$, $3\mathbf{x}$, $\mathbf{x}^T\mathbf{y}$, and $\|\mathbf{x}\|_2$.
2. Normalize $\mathbf{v} = [3, 4]^T$ and verify that the resulting vector has norm 1.

3. Determine whether $[1, 2]^T$ and $[2, 4]^T$ span $\mathbb{R}^2$. Support your answer using rank.
4. Express $[7, 1]^T$ as a linear combination of $[1, 0]^T$ and $[0, 1]^T$.
5. Calculate the cosine similarity between $[4, 3, 0]^T$ and $[5, 2, 1]^T$ using NumPy.
6. Create a dataset with four students and three normalized features. Select a weight vector and calculate all weighted scores using one matrix multiplication.
7. Explain why Euclidean distance can be misleading when features use very different measurement scales.

Key Takeaways

- Vectors represent observations, parameters, directions, and embeddings.
- Vector operations require compatible dimensions and careful shape handling.
- Norm measures length; distance measures separation; cosine similarity compares direction.
- A linear combination forms a new vector by weighting and adding existing vectors.
- Span describes every vector reachable through those linear combinations.
- Observations become points in a feature space, and the representation determines its geometry.
- Matrix-vector multiplication efficiently applies the same weighted rule to many observations.

Key Terms

Feature vector, Vector norm, Unit vector, Euclidean distance, Dot product, Cosine similarity, Linear combination, Span, Feature space, Embedding, Weight vector, Vectorization.